

Urticaria and Angio-oedema

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Introduction:

- Type 1 hypersensitivity reaction.
- Urticaria describes transient pruritic swellings of the skin often referred to as wheals.
- Results from edema in the superficial layers of the skin causing well-demarcated erythematous lesions.
- In most patients no cause can be found (Idiopathic).
- Urticaria is a common skin disorder that affects 20% of the population at some point in their lives.
- Single episode or chronically recurrent.
- Usually self-limiting, controlled by OTC antihistamine.
- Chronic idiopathic urticaria may persist for several years.

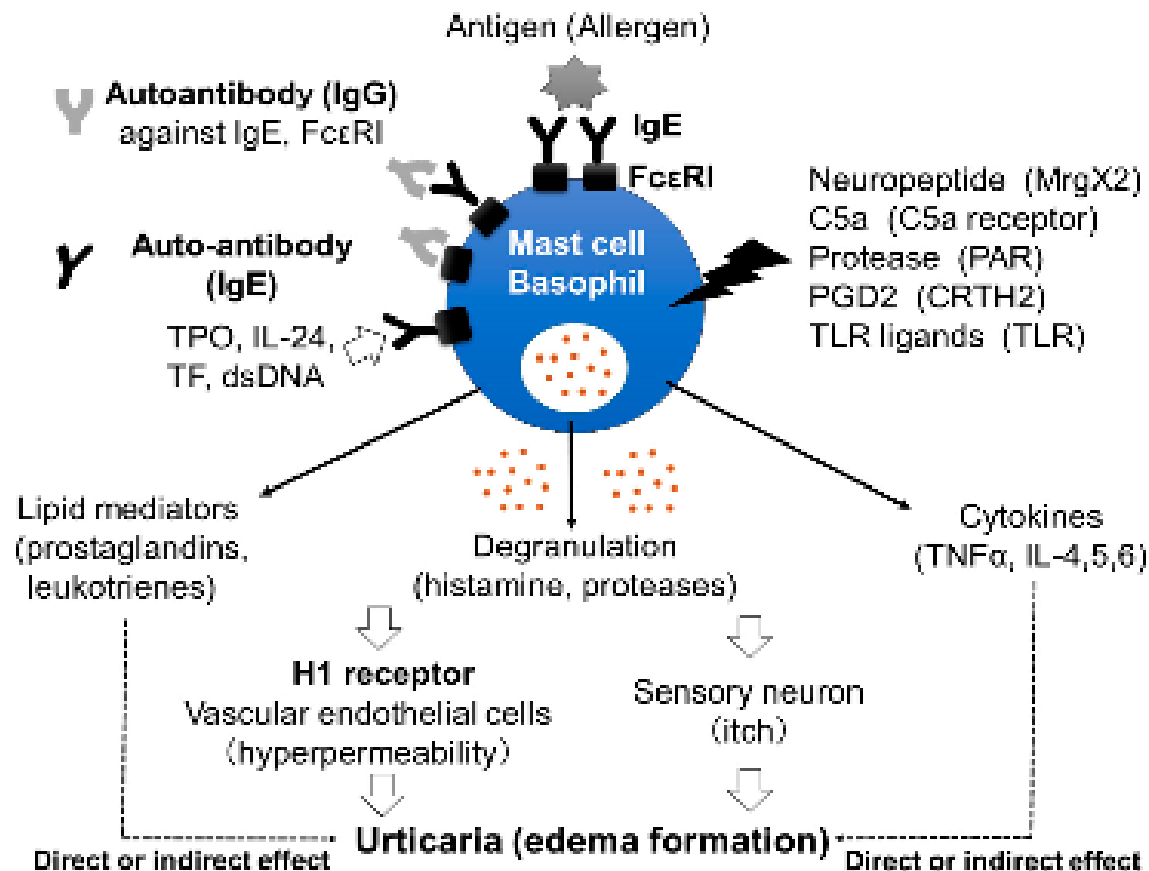
Introduction:

- Angioedema:
 - Painful rather than itchy.
 - Cause diffuse swelling that affects the deeper layers of the skin.
 - Can occur rapidly.
 - May involve the mucocutaneous membranes (Laryngeal oedema is the most serious complication and can be life threatening).

Pathophysiology:

- **Mast cell degranulation** due to a trigger will lead to the release of histamine, bradykinin, and other pro-inflammatory mediators
- This leads to increased **capillary permeability** and **transient leakage of fluids** into surrounding tissue and the development of wheals.
- Urticaria may be IgE mediated with cross-linking of two adjacent IgE receptors, or complement-mediated (causing direct degranulation of mast cells), or mast cells may be directly stimulated by an exogenous or unknown substance.
- In patients with chronic urticaria, histamine can be released spontaneously or in response to non-specific stimuli and their vasculature is more sensitive to histamines.

Pathophysiology:



Clinical History:

- Urticaria is very itchy and angioedema is usually painful.
- Urticaria: itchy spots or rashes lasting minutes or hours (less than 24 h)
- Resolve with no marks on the skin.



Ordinary urticaria

Clinical History:

- BUT, if more than 24 h, painful, or resolves with bruising >> think about urticarial vasculitis (Biopsy is needed to confirm).



Clinical History:

- Patients may feel unwell before the onset of the rash or swelling and rarely in severe reactions anaphylaxis can occur. (You should always ask about any associated respiratory distress).
- An attempt to identify possible **triggers** before the onset of symptoms is important.
- In particular, ask about any food eaten (nausea/vomiting), exercise, heat/cold, sun, medications/infusions, latex exposure, insect stings, animal contact, physical stimuli, infections, family history, and any known medical conditions.

Classification of urticaria:

- Urticaria is traditionally classified as acute or chronic depending on whether symptoms last for less or more than 6 weeks.
- Another approach is to classify urticaria according to the underlying cause, but in 50% of cases no cause is identified (idiopathic).
- Urticaria can be divided into:
Ordinary/idiopathic(acute/chronic), contact allergic, cholinergic, physical or urticarial vasculitis

Urticaria:

Box 5.1 Causes of non-physical urticaria

- Food allergies: fish, eggs, dairy products, nuts and strawberries.
- Food additives: tartrazine dyes and sodium benzoates
- Salicylates: medication and foods.
- Infections: viral, bacterial and protozoal.
- Systemic disorders: autoimmune, connective tissue disease and carcinoma.
- Contact urticaria: meat, fish, vegetables and plants.
- Papular urticaria: persistent urticaria often secondary to insect bites.
- Aeroallergens: pollens, house dust mite and animal dander.

Box 5.2 Causes of physical urticaria

- Heat.
- Sunlight.
- Cold.
- Pressure.
- Water.

Ordinary urticaria:

- Most common form.
- characterized by intermittent fleeting wheals at any skin site, with or without angioedema.
- Lesions may be papular, annular or serpiginous.
- In 50% of patients with ordinary urticaria no underlying cause is found.
- Possible triggers of acute urticaria include infections, vaccinations, medications, and food.
- Generally, the more persistent the attacks of urticaria, the less likely that an underlying cause is identified; this is termed **chronic idiopathic urticaria**.



Annular ordinary urticaria

Cholinergic urticaria:

- Patients are usually aged 10–30 years
- Typically urticaria following a warm shower/bath, or after exercise.
- The lesions consist of pinhead-sized wheals with a red flare around them.
- Avoidance of heat usually helps reduce the frequency and severity of symptoms.



Cholinergic urticaria:

- A rarer form of cholinergic urticaria can result **from exposure to the cold:**
 - Patients report urticaria on exposed skin during cold weather >> lip/tongue/hand swelling following the holding and ingestion of cold beverages, and a more generalized reaction following swimming in an outdoor pool.
 - Affected individuals **should avoid swimming in cold water and ingestion of ice-cold drinks as anaphylaxis and death have been reported.**

Solar urticaria:

- Solar urticaria is a rare condition in which sunlight causes an acute urticarial eruption.
- Patient complains of stinging, burning, and itching at exposed skin sites within 30 min of ultraviolet or artificial light source exposure.
- Lesions resolve rapidly (minutes to hours) when light exposure ceases.
- Management can be difficult but avoidance of sunlight is helpful.



Solar Urticaria ddx:

1. **Photosensitive drug eruptions:** so, a detailed drug history is important.
2. **Porphyria:** lesions resolve with scarring.
3. **Polymorphic light eruption:** lesions take days to weeks to resolve.



Pressure urticaria:

- Urticarial wheals occur at the site of pressure on the skin.
- Around the waistband area (from clothing), soles of feet (from walking), hands (using tools and weight lifting), and buttocks (sitting).
- Occurs immediately or delayed up to 6 h (delayed pressure urticaria), and lesions resolve over several days.



Delayed pressure urticaria

This man had a manual job - by the end of the day his hands and feet were so swollen and painful that he was starting to look for an alternative career

Pressure urticaria:

- Symptoms are usually recurrent over many years.
- Dx confirmed by a pressure challenge test.
- Patients are less likely to respond to antihistamines (patients may respond to **dapsone** or **montelukast**).

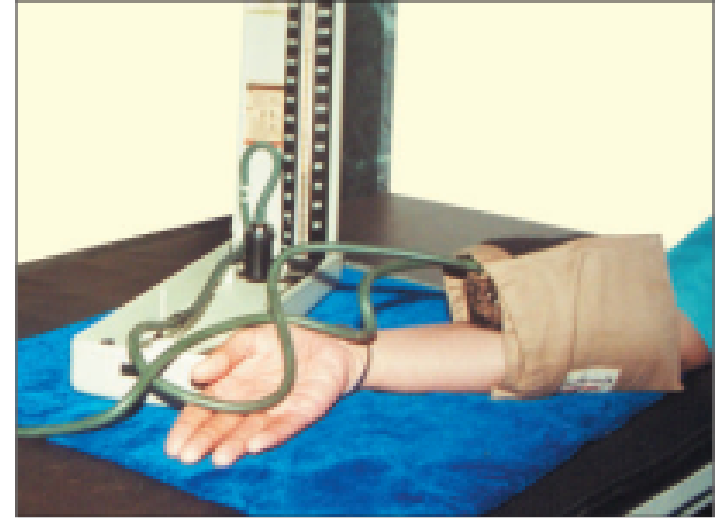


Figure 1: Blood pressure cuff tied with 2 kg weight around right forearm



Angioedema:

- Patients present with swelling +/- urticaria that develops over hours and resolves over days.
- Well-demarcated swelling of the subcutaneous tissues and/or mucous membranes, from increased vascular permeability.
- Allergy to medications may be the cause e.g. ACEI, NSAIDS, bupropion, statins, or PPIs.
- Laryngeal edema is the most serious complication and can be life-threatening (EMERGENCY)

Angioedema of the hand:



Hereditary angioedema:

- Presents in young patients as recurrent severe attacks affecting the skin and mucous membranes
- Tingling, tightness, and pain are the main symptoms associated with edema.
- Patients may suffer associated systemic symptoms (mostly gastrointestinal symptoms) and life-threatening laryngeal edema.
- Caused by a deficiency in C1 (esterase) inhibitor.
- Serum complement C4 levels are low following attacks.
- **Danazol** can be used to reduce the frequency and severity of attacks and **fresh frozen plasma** can be used before elective surgery.

Hereditary angioedema:



General investigations

1. Detailed history and examination.
2. Most patients need no further investigations.
3. If a food allergy is suspected patient can be asked to keep a food diary, particularly if their urticaria is recurrent and episodic.
4. An attempt to elicit dermatographism (exaggerated release of histamine causing wheal and flare) should be made by firmly stroking the skin with a hard object such as the end of a pen; this is usually positive in physical urticaria

Dermatographism:



General investigations:

5- A skin biopsy can be useful if urticarial vasculitis is suspected (painful, > 24 hours, resolves with bruising)

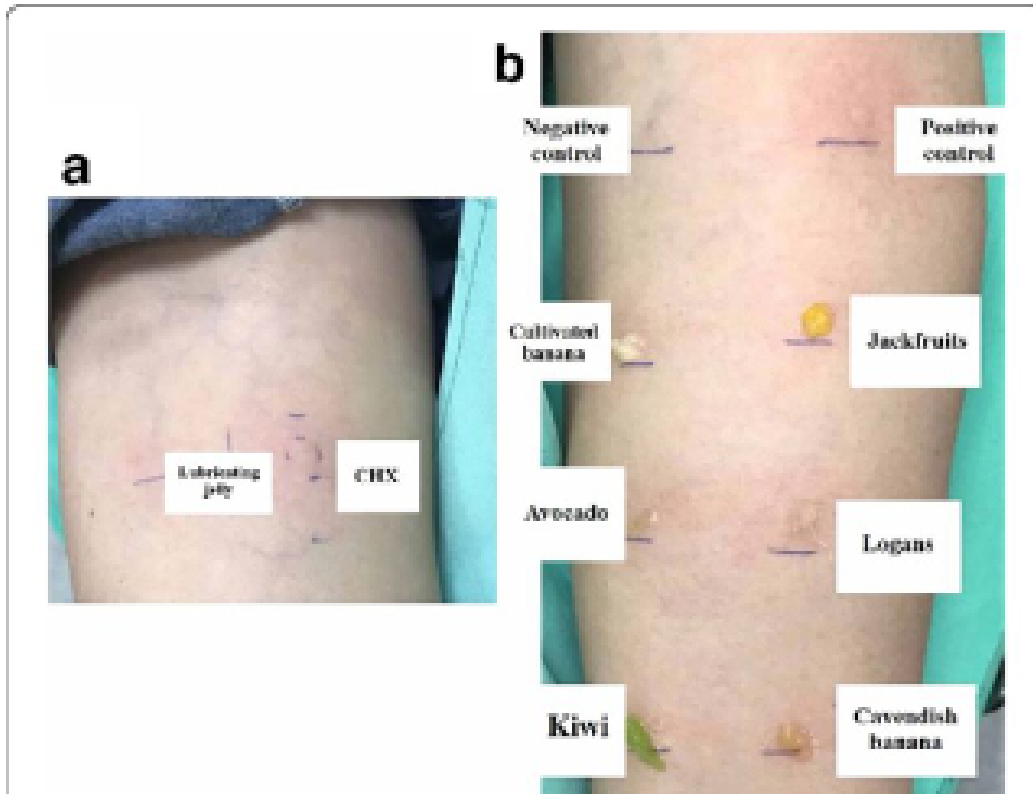
6- In patients with more severe reactions a RAST (radioallergosorbent test); It tests for the amount of specific IgE antibodies in the blood which are present if there is a "true" allergic reaction.

General investigations:

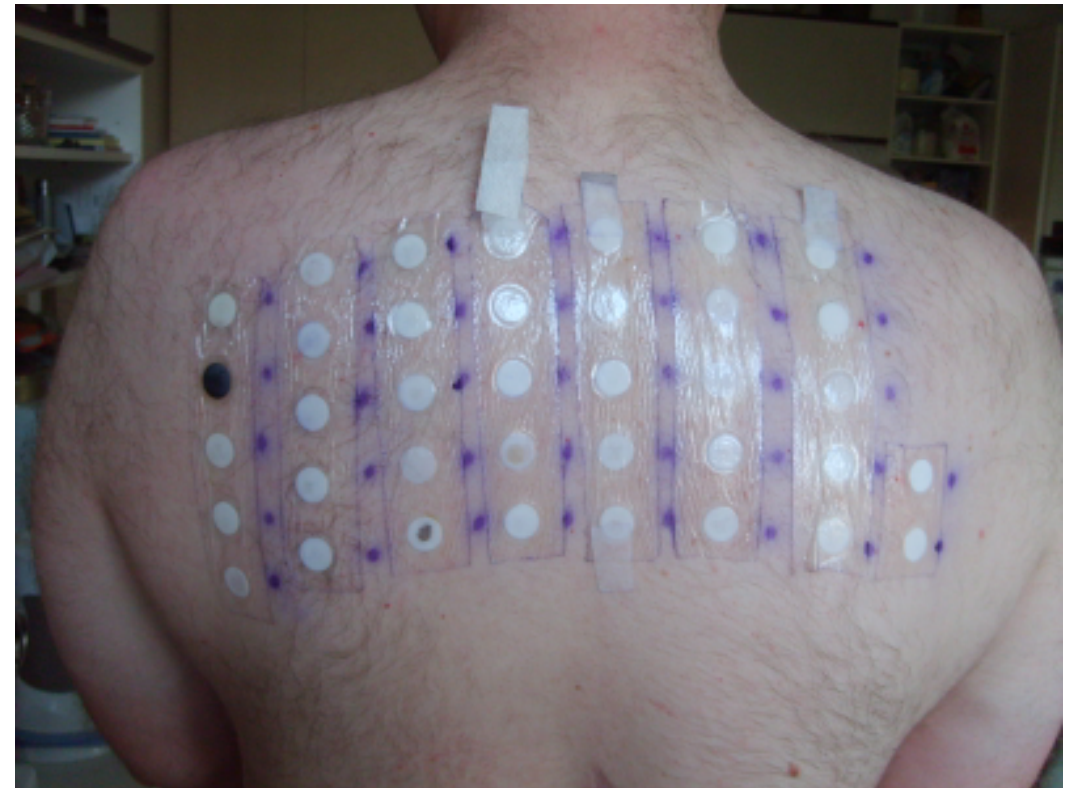
7- Skin prick testing (although not in patients with anaphylaxis) may help identify specific allergies.

8- Patch testing to identify contact urticaria can be undertaken in specialist centers.

Skin prick test	Skin patch test
In hospital settings	Outpatient settings
Test type 1 Hypersensitivity reactions e.g. Urticaria, angioedema	Test type 4 Hypersensitivity reactions e.g. Allergic Contact Dermatitis, or contact urticaria
Risk of anaphylaxis, C.I is hx +ve	No Risk of anaphylaxis



Skin Prick Test



Skin Patch Test

General investigations:

- 9- If you suspect hereditary angioedema check the complement level and C1 esterase level which are usually low.
- 10- If heat or cold are the possible precipitants then exercising for 5min or placing an ice cube on the skin for 20 min may be diagnostic. (respectively)
- 11- Physical urticaria can be elicited by firm pressure on the skin.
- 12- Solar urticaria can be assessed in specialist centers using a solar simulator

General management:

- Patients often make their own observations concerning trigger factors, especially food, medication and insect stings, and know what they should try to avoid.
- Treatment of any underlying medical condition identified should help settle the urticaria.
- **Oral antihistamines are the mainstay** of treatment/prevention of urticaria and angioedema. (2nd generation H1 blockers regular dose, up to 4 tabs daily, not PRN)
e.g. Loratidine, Cetrizine, Desloratidine.

General management:

- In severe cases: synergistic combinations of **H1-receptor blockers** plus H2 blockade (e.g. Famotidine, ranitidine, cimetidine) and **leukotriene receptor antagonists** (e.g. Montelukast, Zafirlukast) are used to control symptoms.
- **Oral corticosteroids** may be indicated in very severe eruptions, particularly those associated with urticarial vasculitis. (not for long duration)

General management:

- Patients who are known to be at risk of severe life-threatening urticaria/ angioedema with respiratory distress may be asked to carry with them a pre-assembled syringe and needle (**EpiPen®**, or **Anapen®**) to inject adrenaline intramuscularly (300–500 µg).
- Omalizumab (a monoclonal antibody that targets IgE) can be used in patients whose urticaria is recalcitrant to other lines of treatment.



Thank You